

Reinsurance Exposure Modeling under Fragmented Policy Data

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Overview

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- Reconstruction of policy-level risk exposure from fragmented database structures
- Integration of:
 - risk components (G , Z)
 - reserves
 - policy attributes and time evolution
- Objective:
 - derive consistent exposure representation
 - apply retention-based allocation for reinsurance

Core Modeling Problem

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- No explicit data model for:
 - current policy state
 - time evolution of exposure
- Multiple inconsistent sources:
 - versioned tables (historical vs. actual)
 - missing or misaligned timestamps
- Required:
 - reconstruction of state
 - definition of exposure per policy
 - aggregation at client level

Data Representation

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- Risk components:
 - G-type (main coverage)
 - Z-type (additional riders)
- Financial quantities:
 - sum at risk
 - reserves (gross / net)
 - written premium
- Time reference:
 - booking timeline as primary axis
 - indirect linkage to policy states

Exposure Transformation

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- Adjustment of main risk component:

$$G_{\text{adj}} = \text{Sum at risk} - \text{Reserve}$$

- Rider risks:
 - aggregated across multiple entries
 - dependent on risk type classification
- Result:
 - unified exposure measure per policy

Aggregation Logic

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- Exposure aggregated at client level
- Multiple policies per client:
 - multiple G-components possible
 - non-trivial ordering required
- Output:
 - ordered exposure vector per client
 - basis for allocation algorithm

Retention-Based Allocation

- Retention threshold R (e.g. 15000)
- Let k denote the number of exposure components
- Let G_i denote individual exposure components ($i = 1, \dots, k$) and

$$S_k = \sum_{i=1}^k G_i$$

- Sequential allocation based on cumulative exposure:

$$\text{Ceded}_k = \max(0, S_k - R) - \max(0, S_{k-1} - R)$$

- Piecewise allocation of exposure above the retention threshold

Time Dimension

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- Exposure evaluated over booking timeline
- Issues:
 - missing timestamps in core tables
 - multiple versions per date
- Resolution:
 - selection of representative version
 - alignment via booking records

Implementation Aspects

- SQL-based pipeline:
 - views for exposure reconstruction
 - stored procedures for allocation logic
- Output structures:
 - exposure over time
 - current reinsurance allocation
 - derived premium quantities
- iterative validation against source data

Engineering Challenges

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- Missing or inconsistent join keys
- Multiple representations of the same risk
- Data inconsistencies:
 - duplicated timestamps
 - incomplete reserve history
- Necessity of implicit assumptions:
 - selection rules
 - aggregation conventions

What was actually solved

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- Reconstruction of a consistent exposure model from fragmented data
- Definition of transformation from raw risk to effective exposure
- Formalization of retention-based allocation algorithm
- Creation of reproducible pipeline for reinsurance inputs
- Delivery of production-ready implementation validated on real portfolio data

Outcome

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- Consistent exposure representation across policies and time
- Deterministic allocation of risk above retention threshold
- Structured outputs usable for:
 - reinsurance reporting
 - downstream actuarial calculations
- Deployment of SQL procedures into production environment
- Validation of results directly within operational system against business expectations